

Rahul Singh

Software Engineer | Noida, India | +91-9752079591 | rahul.singh.cse79@gmail.com
linkedin.com/in/rahul-singh-a32261206 | github.com/Rahulcse79 | leetcode.com/u/rahulcse79 | ai-voice-portfolio.onrender.com

PROFESSIONAL SUMMARY

Software Engineer with 2.5+ years building scalable, low-latency distributed systems in **Go (Golang), TypeScript, Java, and C/C++** serving 50K+ daily requests. Expertise in **microservices, REST APIs, real-time WebSocket streaming, AI/ML integration, Docker, Kubernetes, PostgreSQL, Redis** on AWS and GCP. Delivers 40% latency cuts, 99.9% uptime, and 70% faster CI/CD releases.

PROFESSIONAL EXPERIENCE

Software Engineer | Coral Telecom

Dec 2023 – Present

Go · TypeScript · Java · Spring Boot · Node.js · PostgreSQL · Redis · Docker · GitLab CI/CD · WebSockets

- Architected **12+ RESTful microservices** in **Go, Node.js, Spring Boot** serving **50K+ daily API requests** at **99.9% uptime** across distributed production environments.
- Reduced API latency **40%** via **Redis caching, PostgreSQL** query plan optimization, and connection pooling across high-throughput services.
- Engineered **CI/CD pipelines** on **Docker and GitLab CI**, cutting deployment time **70%** and enabling **zero-downtime releases** to production.
- Delivered real-time modules using **WebSockets and Server-Sent Events**, powering sub-second streaming for enterprise monitoring dashboards across 20+ tenants.
- Authored unit and integration tests with **Jest and JUnit**, achieving **85%+ code coverage** and reducing regression incidents by 45%.
- Led **50+ code reviews** enforcing **SOLID** and clean architecture, cutting post-release defects **30%**; mentored **3 junior engineers** (2-week faster onboarding).

PROJECTS

AI-Powered Real-Time IVR Voice Agent

C/C++ · FreeSWITCH · libwebsockets · GCP STT/TTS · PostgreSQL

- Engineered a production **FreeSWITCH C/C++ module** streaming live call audio over **WebSockets** to an AI backend at **sub-200ms end-to-end latency**.
- Integrated **Google Cloud STT/TTS** to autonomously resolve **80%+ customer queries**, cutting human agent workload by **60%**.
- Designed an **L16 audio pipeline** (dual-channel buffering, 8/16 kHz resampling, WAV archival) with **PostgreSQL/libpq** persistence and auto-migration.
- Sustained **100+ concurrent WebSocket connections** via mutex-guarded circular buffers and RAII; **zero memory leaks** verified by **Valgrind**.

Dressly — Full-Stack E-Commerce Platform

React.js · Node.js · Express · MongoDB · JWT · Stripe · Cloudinary

- Built end-to-end platform with **React.js** storefront, admin dashboard, and **Node.js/Express** backend in clean **MVC** architecture across 3 environments.
- Designed **20+ REST endpoints** with **JWT + bcrypt auth**, cart, orders, reviews, and **RBAC** admin controls supporting 1K+ concurrent sessions.
- Integrated **Stripe** (PCI-compliant checkout) and **Cloudinary** CDN, improving page load **35%** and achieving a **95+ Lighthouse** score.

TECHNICAL SKILLS

- **Languages:** Go (Golang), Java, C/C++, TypeScript, JavaScript (ES6+), Python, SQL
- **Frameworks:** Spring Boot, Node.js, Express.js, React.js, Next.js, GraphQL, REST APIs, Microservices
- **Databases & Cache:** PostgreSQL, MySQL, MongoDB, Redis
- **Cloud & DevOps:** Docker, Kubernetes (K8s), AWS (EC2, S3), GCP, GitLab CI/CD, GitHub Actions, Nginx, Linux
- **Tools & Integrations:** Git, WebSockets, OpenAI APIs, Google Cloud STT/TTS, Stripe, Postman, Maven, NPM
- **Testing & Concepts:** Jest, JUnit, Valgrind, Unit/Integration Testing; Distributed Systems, System Design, SOLID, Design Patterns, Multithreading, Real-Time Streaming, Observability, Agile/Scrum, AI Integration

EDUCATION

B.Tech, Computer Science & Engineering, NRI Institute of Science and Technology, Bhopal

2019 – 2023 | **CGPA:**

8.0/10

ACHIEVEMENTS

- Solved **500+ LeetCode problems** across DSA and system design; ranked **top 8% globally** with consistent daily streak.
- Recognized as **top-performing engineer** at Coral Telecom; delivered AI-IVR module **2 weeks ahead of schedule**.
- **Udemy Certified:** Mastering DSA in C++ (trees, graphs, DP, greedy); **LeetCode DCC Jan 2023 Badge** (31/31 daily challenges).